1 Case Study 1: CRISPR Field Review (2010–2024)

1.1 Objective

Map the structure, growth, and evolution of CRISPR gene editing research from its emergence to its current state using open bibliometric data.

1.2 Setup

```r
library(tidyverse)
library(openalexR)
library(igraph)
library(tidygraph)
library(ggraph)
library(quanteda)
library(tidytext)
library(glue)
library(gt)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

1.3 Data acquisition

```r
works <- oa_fetch(
  entity = "works",
  search = "CRISPR",
  from_publication_date = "2010-01-01",
  to_publication_date = "2024-06-30",
  type = "article",
  options = list(sample = 500, seed = 42)
)

works <- works |>
  mutate(year = year(publication_date))

cat(glue("CRISPR articles retrieved: {nrow(works)}\n"))
```

```
#> CRISPR articles retrieved: 500
```

```r
cat(glue("Year range: {min(works$year)}--{max(works$year)}\n"))
```

```
#> Year range: 2010--2024
```

1.4 Publication growth

```
works |>
  count(year) |>
  ggplot(aes(x = year, y = n)) +
  geom_col(fill = palette_sci(1)) +
  labs(x = "Year", y = "Publications") +
  theme_sci()
```

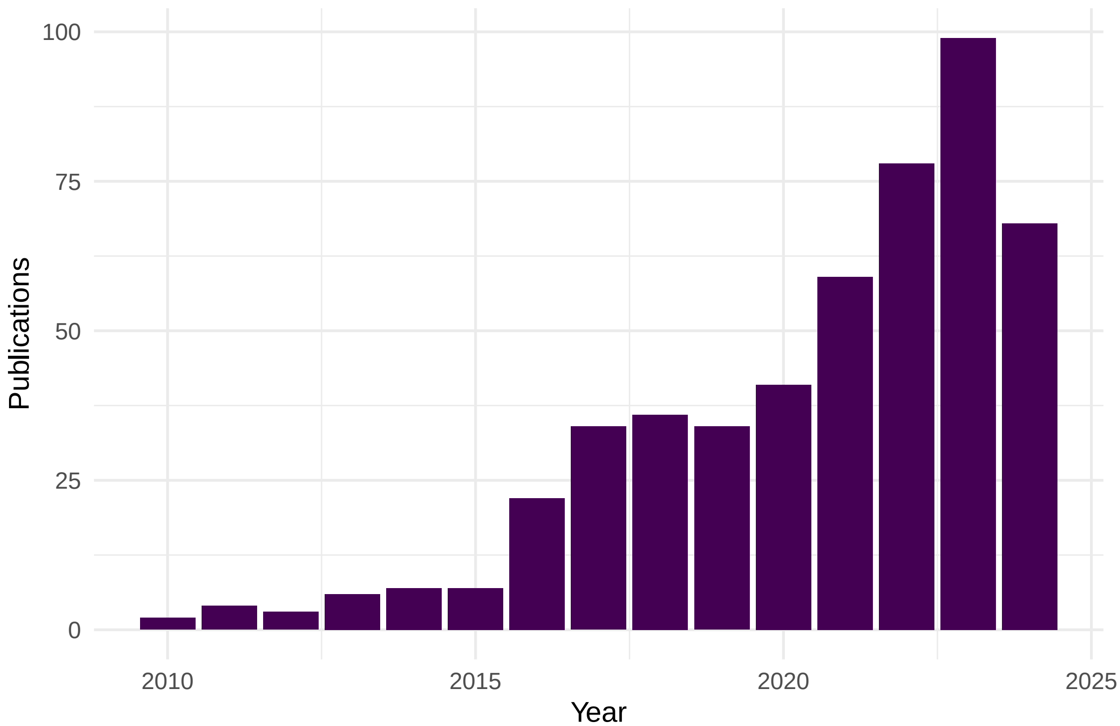

Figure 1: Annual publication output in the CRISPR field.

1.5 Citation landscape

```
ggplot(works, aes(x = cited_by_count)) +
  geom_histogram(binwidth = 10, fill = palette_sci(1), colour = "white") +
  labs(x = "Citations", y = "Papers") +
  theme_sci()
```

```
works |>
  arrange(desc(cited_by_count)) |>
  head(10) |>
  select(display_name, year, cited_by_count, source_display_name) |>
  gt()
```

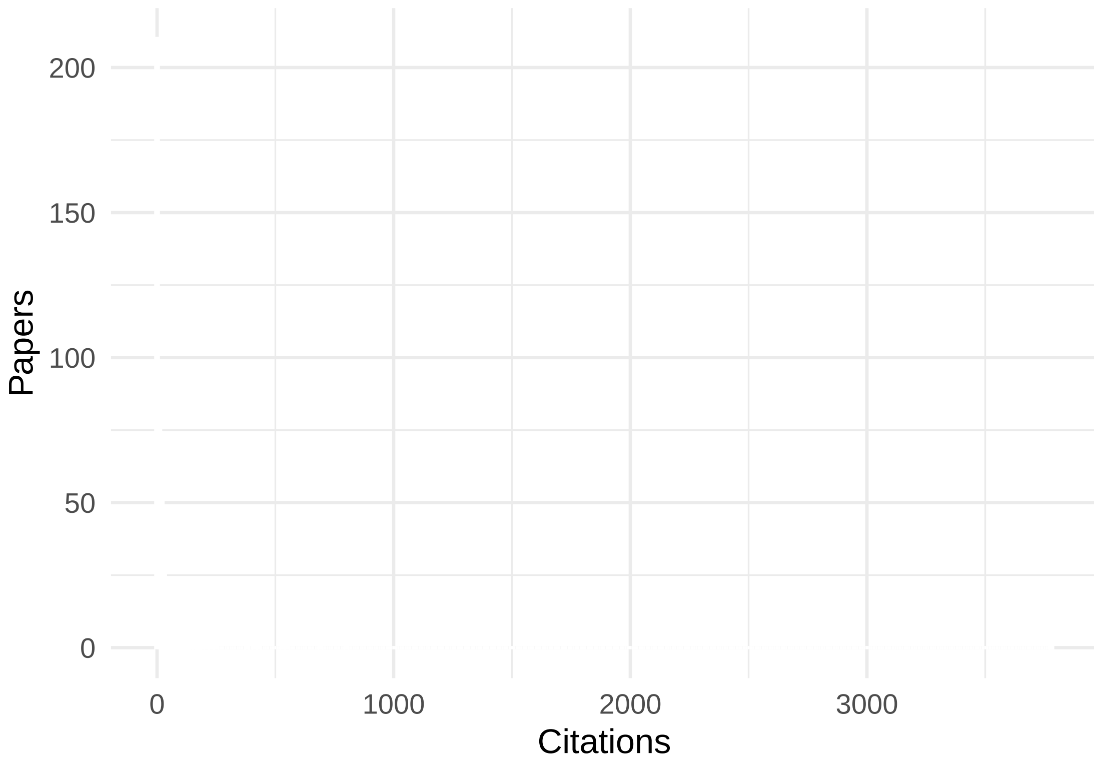

Figure 2: Citation distribution of CRISPR articles.

1.6 Co-authorship network

```
author_data <- works |>
  select(id, authorships) |>
  unnest(authorships, names_sep = "_") |>
  select(work_id = id, author_id = authorships_id,
         author_name = authorships_display_name) |>
  filter(!is.na(author_id))

edges <- author_data |>
  inner_join(author_data, by = "work_id", suffix = c("_1", "_2"),
             relationship = "many-to-many") |>
  filter(author_id_1 < author_id_2) |>
  count(author_id_1, author_id_2, name = "weight")

g <- graph_from_data_frame(
  edges |> select(author_id_1, author_id_2, weight),
  directed = FALSE
) |> simplify(edge.attr.comb = list(weight = "sum"))

comp <- components(g)
giant <- induced_subgraph(g, which(comp$membership == which.max(comp$csize)))
V(giant)$community <- as.factor(membership(
  cluster_leiden(giant, resolution_parameter = 1.0,
                 objective_function = "modularity")
```

display_name
Nucleic acid detection with CRISPR-Cas13a/C2c2
A high-fidelity Cas9 mutant delivered as a ribonucleoprotein complex enables efficient gene editing in hun
Efficient genome engineering in human pluripotent stem cells using Cas9 from <i>Neisseria meningitidis<
Gene targeting by the <scp>TAL</scp> effector PthXo2 reveals cryptic resistance gene for bacterial bli
An enhanced CRISPR repressor for targeted mammalian gene regulation
Efficient targeted multiallelic mutagenesis in tetraploid potato (Solanum tuberosum) by transient CRISP
Programmable Removal of Bacterial Strains by Use of Genome-Targeting CRISPR-Cas Systems
Optimized gene editing technology for <i>Drosophila melanogaster</i> using germ line-specific Cas9
Liver-Derived Signals Sequentially Reprogram Myeloid Enhancers to Initiate and Maintain Kupffer Cell I
RAP2 mediates mechanoresponses of the Hippo pathway

```r
))
V(giant)$degree <- degree(giant)

cat(glue("Network: {vcount(giant)} nodes, {ecount(giant)} edges\n"))
```

```
#> Network: 52 nodes, 426 edges
```

```r
set.seed(42)
ggraph(as_tbl_graph(giant), layout = "fr") +
  geom_edge_link(alpha = 0.1, colour = "grey60") +
  geom_node_point(aes(size = degree, colour = community), alpha = 0.7) +
  scale_size_continuous(range = c(0.5, 5), guide = "none") +
  scale_colour_manual(values = palette_sci(n_distinct(V(giant)$community))) +
  labs(colour = "Community") +
  theme_void(base_family = "sans", base_size = 11) + theme(legend.position = "bottom")
```

1.7 Topic evolution

```r
text_df <- works |>
  filter(!is.na(abstract), nchar(abstract) > 50) |>
  transmute(doc_id = id, text = paste(display_name, abstract, sep = ". "), year)

corp <- corpus(text_df, docid_field = "doc_id", text_field = "text")
toks <- tokens(corp, remove_punct = TRUE, remove_numbers = TRUE) |>
  tokens_tolower() |>
  tokens_remove(stopwords("en")) |>
  tokens_remove(c("study", "paper", "results", "using", "based"))

dfmat <- dfm(toks) |> dfm_trim(min_termfreq = 5, min_docfreq = 3)

top_by_year <- map_dfr(unique(text_df$year), function(yr) {
  docs <- docvars(dfmat, "year") == yr
  if (sum(docs) < 5) return(tibble())
  top <- topfeatures(dfmat[docs, ], 5)
```

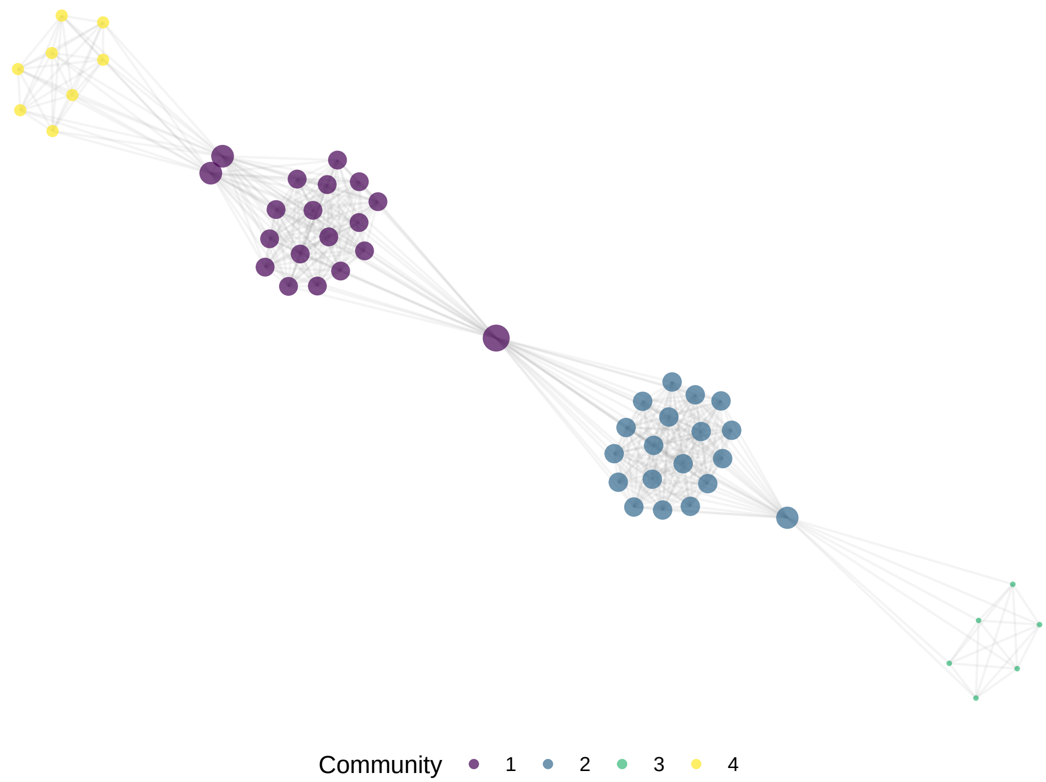

Figure 3: Co-authorship network of CRISPR researchers.

```
  tibble(year = yr, term = names(top), freq = unname(top))
})
```

```
top_by_year |>
  group_by(year) |>
  mutate(term = reorder_within(term, freq, year)) |>
  ggplot(aes(x = freq, y = term)) +
  geom_col(fill = palette_sci(1)) +
  facet_wrap(~ year, scales = "free_y", ncol = 4) +
  scale_y_reordered() +
  labs(x = "Frequency", y = NULL) +
  theme_sci(base_size = 8)
```

1.8 Key findings

1. **Explosive growth**: CRISPR publications grew exponentially from 2012, reflecting the rapid adoption of Cas9-based editing.
2. **Citation concentration**: A small number of foundational papers dominate the citation landscape.
3. **Collaborative structure**: The co-authorship network shows distinct communities, likely corresponding to different application domains (therapeutics, agriculture, basic biology).
4. **Topic evolution**: Early terms focus on methodology; later years shift toward applications and clinical translation.

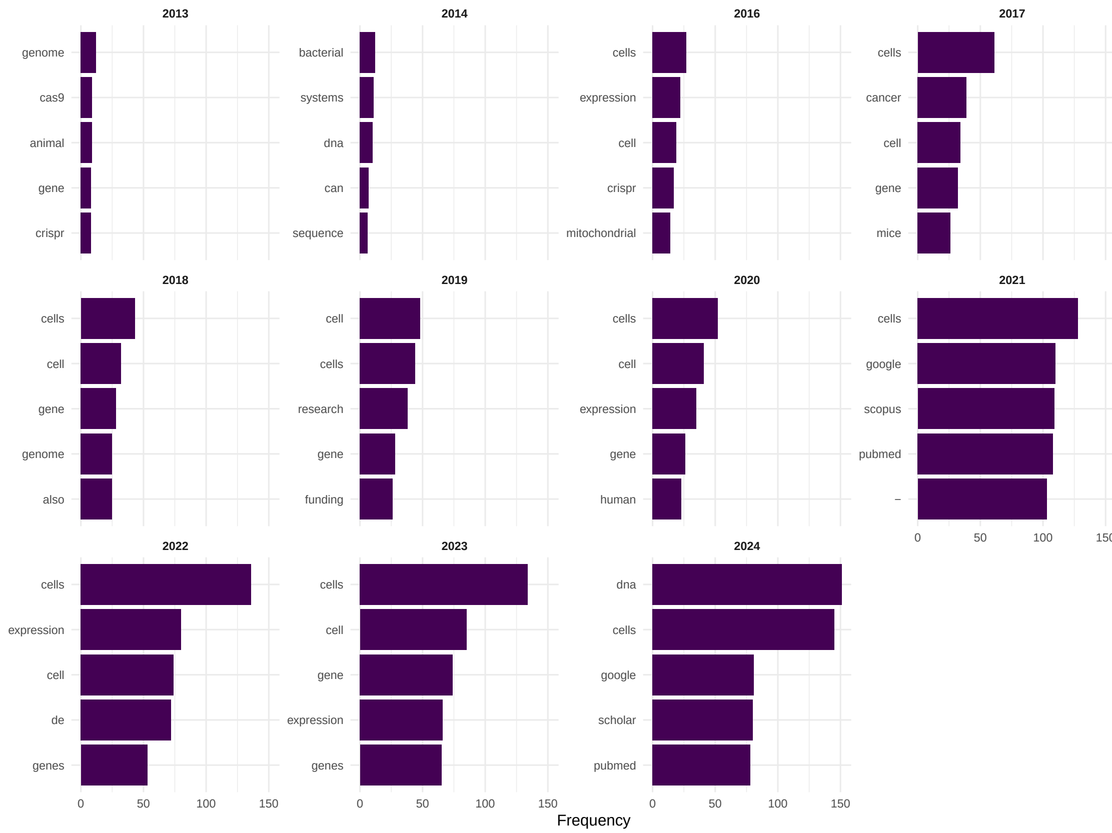

Figure 4: Top terms by year showing topical evolution.

1.9 Lessons learned

- OpenAlex sampling provides a representative snapshot but may miss some highly specialised or non-English publications.
- The citation distribution is extreme: median citations are far below the mean, making median-based statistics essential.
- Co-authorship networks in fast-growing fields are fragmented; many research groups work independently.